

Minwoo James Kim

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Research Interests

Quantum machine learning; quantum information theory; expressivity theory of quantum algorithms; quantum reservoir computing; quantum dynamics; quantum computer architecture.

Education

Seoul National University, Computer Science & Engineering Seoul, Korea
M.S. in Computer Science & Engineering, 2025–present (expected Feb. 2027)

- Quantum Information and Quantum Computing Lab; advisor Prof. Taehyun Kim.

Seoul National University, Physics & Astronomy Seoul, Korea
B.S. in Physics, 2019–2025; GPA 3.53/4.00

- Bachelor's thesis, *Practical Limitations on Training of Variational Quantum-Neural Hybrid Eigensolver*.

Publications and Preprints

- **M. J. Kim**, K. K. Park, K. Lee, J. Bang, and T. Kim. *A Rigorous Hybridization of Variational Quantum Eigensolver with Classical Neural Network*. arXiv:2602.17295 (2026); manuscript under revision at *Quantum Science and Technology*.
- K. K. Park, K. Choi, **M. J. Kim**, G. Song, and T. Kim. *Quantum Linear Multistep Method for Using a Quantum Oracle with Differential Equations*. arXiv:2501.03781 (2025).
- **M. J. Kim**, J. Bang, and T. Kim. *Power of Quantum Reservoir Computing*. Manuscript in preparation.
- **M. J. Kim**, and T. Kim. *Prethermal-Floquet Quantum Reservoir Computing with Tunable Memory and Measurement-Efficient Readout*. Manuscript in preparation.

Conference Proceedings

- **M. J. Kim**, K. K. Park, B. Cho, and T. Kim. *Architecture of a Modular Operating System for High-Level Quantum Computation*. *IEEE Quantum Week (QCE) Proceedings*, 2025, pp. 430–431. doi:10.1109/QCE65121.2025.10380.
- K. K. Park, B. Cho, **M. J. Kim**, and T. Kim. *Quantum Circuit Compilation for Small Scale Trapped Ion Quantum Computer*. *IEEE Quantum Week (QCE) Proceedings*, 2025, pp. 432–433. doi:10.1109/QCE65121.2025.10381.

Selected Presentations and Conference Abstracts

- **Oral presentation**, Korean Physical Society Fall Meeting, 2026. *Quantum Reservoir Computing Using Prethermal Floquet Dynamics*.
- **Poster presentation**, Korean Physical Society Fall Meeting, 2026. *Floquet-Engineered Persistent Memory and Efficient Readout in Quantum Reservoir Computing*.
- **Poster presentation**, AQIS 2026. *A Rigorous Hybridization of Variational Quantum Eigensolver and Classical Neural Network*.
- **Oral presentation**, QISK Annual Meeting, 2026. *A rigorous hybridization of variational quantum eigensolver and classical neural network*.
- **Poster presentation**, IEEE Quantum Week (QCE), 2025. *Architecture of a Modular Operating System for High-Level Quantum Computation*.
- **Poster presentation**, IEEE Quantum Week (QCE), 2025. *Quantum Circuit Compilation for Small Scale Trapped Ion Quantum Computer*.
- **Oral presentation**, QISK Annual Meeting, 2025. *Practical Limitations on Training of Variational Quantum-Neural Hybrid Eigensolver*.

Selected Software

- **vqnhelib** (github.com/miinukim/vqnhelib) — numerical tools for quantum-neural hybrid eigensolvers.
- **pyqres** (github.com/miinukim/pyqres) — reservoir simulations and expressivity diagnostics.
- **pyqcsnu** (github.com/snu-quiqcl/pyqcsnu) — Qiskit-compatible interface for circuit submission and laboratory results.
- **maqpie** (github.com/snu-quiqcl/maqpie) — Graphical user interface for unified control of trapped-ion experiments.

Research Experience

Quantum AI for automated driving (Hyundai Motor Group) 2023–2024

- Explored quantum machine learning approaches to automated driving in a joint research project.
- Combined quantum edge detection with Faster–RCNN for a fast object detection algorithm.

Quantum-neural hybrid variational algorithms 2024–2026

- Participated in IonQ research mentorship program, investigated error mitigation capabilities of neural post–processing using convolutional neural networks.
- Identified finite-shot limits of diagonal neural reweighting, developed a unitary phase-based alternative that preserves the variational bound, and tested it on various spin models.

Quantum reservoir computing 2026–present

- Developed an operator-valued Volterra framework and numerical diagnostics to identify temporal features accessible through measurement.
- Designed a local, prethermal Floquet reservoir for classical-shadow readout and analyzed the tradeoff between memory persistence and feature visibility.

Software architecture for trapped-ion quantum computers 2025–2026

- Developed software linking circuit submission, compilation, scheduling, hardware dispatch, and result processing for laboratory trapped-ion hardware.
- Built a Qiskit-compatible client and contributed to experiment-control software used in the QuIQCL laboratory.

Honors and Awards

- Presidential Science Scholarship, Korea Student Aid Foundation 2019–2024
- Qiskit Advocate 2025–present
- IBM Quantum Special Award, QHackathon 2024 2024
- Quantum Excellence, Qiskit Global Summer School 2023
- Center Director Award, QHackathon 2023 2023

Patent Application

T. Kim and **M. J. Kim**. *Methods and Apparatus for Resource-Efficient Calculation of Ground State of Quantum Systems, and Recording Medium Thereof.* Application filed by Seoul National University R&DB Foundation, 2025 (SNU202505234).

Teaching and Leadership

- Teaching Assistant, *Topics in New Computer Technology*, Seoul National University Fall 2026
- Teaching Assistant, *Physics Experiment*, Seoul National University Spring 2024
- Undergraduate physics mentor, Seoul National University Spring 2024
- President, Seoul National University Piano Club 2020
- Team captain, Seoul National University College of Human Ecology Baseball Team 2026

Technical Skills

Python, Java, OCaml; Qiskit, PennyLane, PyTorch, NumPy; Linux, LaTeX, Django, FastAPI.
Languages — Korean (native), English (fluent; TOEFL iBT 110/120).